

xdisturb REST API Specifications (v1)

Mobile API User Documentation & Integration Guide

Document ID:	API-2026-XDISTURB-V1	API Version:	v1.0
Base URL (Production):	https://api.xdisturb.et:8003/api/v1	Auth Mechanism:	Bearer Token (Firebase / JWT)
Format:	JSON (application/json)	Audit Evaluation:	Hibir Audit (audit.hibir.et) — July 30, 2026

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Authentication Mechanism: Bearer Token (Authorization: Bearer <FirebaseIdToken / JWT>)

Data Format: JSON (Content-Type: application/json)

Audit Target: Hibir Audit (audit.hibir.et) — Audit Date: July 30, 2026

Date: August 10, 2026

Document Status: Official Specification

Document Control & Versioning

Version	Date	Author / Role	Summary of Changes	Approved By
v1.0.0	July 30, 2026	API Architecture Working Group	Initial v1 REST API Integration Guide & Schema Definitions	Technical Director
v1.0.1	August 10, 2026	Lead Backend Engineer	Added SSL Pinning Hashes & Chapa Webhook Specifications	Head of Engineering

1. Overview & General Architecture

The **xdisturb REST API** supports native mobile application clients (Android & iOS) for user registration, SMS OTP dispatch, verification, silent zone retrieval, and Chapa mobile payment initialization.

Key Architectural Guidelines:

- Protocol & TLS Enforcement:** All API requests must use **HTTPS (TLS 1.2 or TLS 1.3)** over port **8003** . Cleartext HTTP is rejected at the network firewall level.

- **Content Negotiation:** All request payloads must include `Content-Type: application/json`. All responses return JSON payloads formatted UTF-8.
- **Authentication:** Secured endpoints require a valid Firebase ID Token or JWT passed in the HTTP Authorization header:

```
Authorization: Bearer <FirebaseIdToken>
```



Syntax error in text
mermaid version 10.9.8

2. Authentication Endpoints

2.1 Request OTP: `POST /auth/send-otp`

Sends a 6-digit SMS verification code to a valid Ethiopian phone number.

- **Full Endpoint URL:** `https://api.xdisturb.et:8003/api/v1/auth/send-otp`
- **HTTP Method:** `POST`
- **Authentication:** None (Public Endpoint)
- **Headers:** `Content-Type: application/json`
- **Request Payload:**

```
{
  "phone": "+251911223344"
}
```

- **Response (200 OK):**

```
{
  "status": "success",
  "message": "OTP code dispatched via SMS"
}
```

- **Rate Limit:** Maximum 3 OTP requests per phone number within a 15-minute window. Rate-limited requests return `HTTP 429 Too Many Requests`.

2.2 Verify OTP: `POST /auth/verify-otp`

Verifies the SMS OTP code and returns a custom Firebase authorization token.

- **Full Endpoint URL:** `https://api.xdisturb.et:8003/api/v1/auth/verify-otp`

- **HTTP Method:** POST
- **Headers:**
 - Content-Type: application/json
 - Authorization: Bearer <OptionalAnonymousToken>
- **Request Payload:**

```
{
  "phone": "+251911223344",
  "code": "849201"
}
```

- **Response (200 OK):**

```
{
  "status": "success",
  "customToken": "eyJhbGciOiJSUzI1NiIsIm..."
}
```

- **Error Response (400 Bad Request):**

```
{
  "status": "error",
  "code": "INVALID_OTP",
  "message": "The 6-digit code provided is invalid or expired."
}
```

3. Payments & Subscription Endpoints

3.1 Initiate Payment Intent: POST /payments/initiate

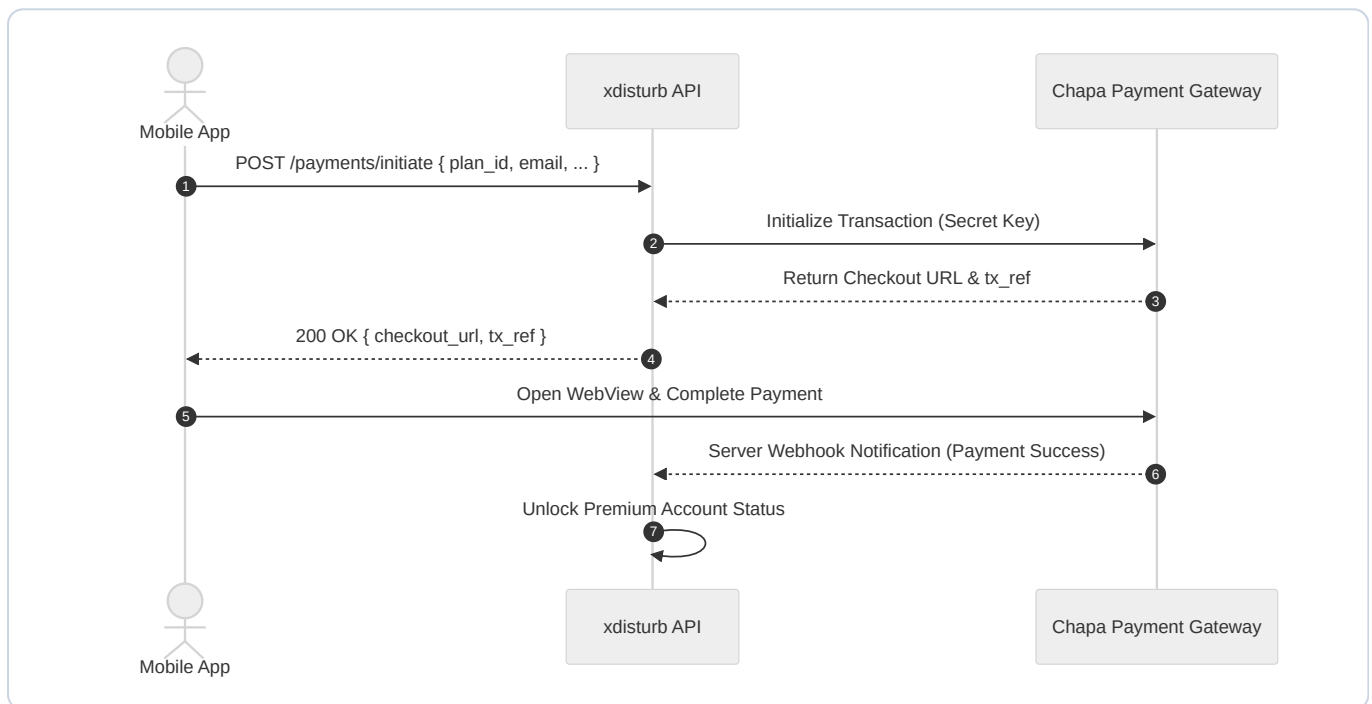
Generates a **Chapa Checkout** webview transaction URL for purchasing premium subscriptions via Telebirr, CBE Birr, or Bank Cards.

- **Full Endpoint URL:** https://api.xdisturb.et:8003/api/v1/payments/initiate
- **HTTP Method:** POST
- **Headers:**
 - Content-Type: application/json
 - Authorization: Bearer <FirebaseIdToken>
- **Request Payload:**

```
{
  "plan_id": "premium_monthly",
  "first_name": "Abebe",
  "last_name": "Bikila",
  "email": "abebe@example.com"
}
```

- **Response (200 OK):**

```
{
  "status": "success",
  "checkout_url": "https://checkout.chapa.co/checkout/web/pay/tx_9841270",
  "tx_ref": "tx_9841270"
}
```



3.2 Verify Payment Status: `GET /payments/verify/{tx_ref}`

Checks the completion status of a Chapa payment transaction.

- **Full Endpoint URL:** `https://api.xdisturb.et:8003/api/v1/payments/verify/tx_9841270`
- **HTTP Method:** `GET`
- **Headers:** `Authorization: Bearer <FirebaseIdToken>`
- **Response (200 OK):**

```
{
  "status": "success",
  "data": {
    "tx_ref": "tx_9841270",
    "amount": 150.00,
    "currency": "ETB",
    "payment_status": "success",
    "paid_at": "2026-08-10T11:55:00Z"
  }
}
```

4. Silent Zone Data Model & Endpoints

4.1 Silent Zone JSON Data Model

The canonical JSON representation of a Silent Zone entity:

```
{
  "id": "sz_addis_al_anwar",
  "name": "Al-Anwar Mosque",
  "center": {
    "latitude": 9.0300,
    "longitude": 38.7400
  },
  "radius": 150.0,
  "type": "mosque",
  "isActive": true
}
```

Field Specifications:

Field Name	Type	Description	Allowed Values / Constraints
id	String	Unique zone identifier	Prefix <code>sz_</code> followed by alphanumeric string
name	String	User-facing display name	3 to 64 characters
center.latitude	Double	WGS84 Latitude coordinate	-90.0 to +90.0
center.longitude	Double	WGS84 Longitude coordinate	-180.0 to +180.0
radius	Double	Geofence coverage radius (meters)	50.0 to 500.0
type	String	Category classification	"mosque" , "church" , "library" , "custom"
isActive	Boolean	Operational monitoring status	true (Active), false (Disabled)

4.2 Query Nearby Silent Zones: GET /zones/nearby

Retrieves pre-populated and user custom Silent Zones within a given radius of a geographic coordinate.

- **Full Endpoint URL:** `https://api.xdisturb.et:8003/api/v1/zones/nearby?lat=9.0300&lng=38.7400&radius=2000`
- **HTTP Method:** GET
- **Headers:** Authorization: Bearer <FirebaseIdToken>
- **Query Parameters:**
 - `lat` (required, double): User latitude
 - `lng` (required, double): User longitude
 - `radius` (optional, integer): Search radius in meters (default 2000 , max 10000)
- **Response (200 OK):**

```
{
  "status": "success",
  "count": 2,
  "data": [
    {
      "id": "sz_addis_al_anwar",
      "name": "Al-Anwar Mosque",
      "center": { "latitude": 9.0300, "longitude": 38.7400 },
      "radius": 150.0,
      "type": "mosque",
      "isActive": true
    },
    {
      "id": "sz_custom_office_hq",
      "name": "HQ Office",
      "center": { "latitude": 9.0320, "longitude": 38.7450 },
      "radius": 100.0,
      "type": "custom",
      "isActive": true
    }
  ]
}
```

4.3 Create Custom Silent Zone: **POST** /zones/custom

Creates a user-defined custom Silent Zone.

- **Full Endpoint URL:** `https://api.xdisturb.et:8003/api/v1/zones/custom`
- **HTTP Method:** `POST`
- **Headers:**
 - `Content-Type: application/json`
 - `Authorization: Bearer <FirebaseIdToken>`
- **Request Payload:**

```
{
  "name": "HQ Meeting Room",
  "center": {
    "latitude": 9.0100,
    "longitude": 38.7500
  },
  "radius": 100.0,
  "type": "custom"
}
```

• **Response (201 Created):**

```
{
  "status": "success",
  "data": {
    "id": "sz_custom_981240",
    "name": "HQ Meeting Room",
    "center": { "latitude": 9.0100, "longitude": 38.7500 },
    "radius": 100.0,
    "type": "custom",
    "isActive": true
  }
}
```

5. Standard Error Handling & HTTP Status Codes

All API errors return a uniform JSON error payload:

```
{
  "status": "error",
  "code": "EXPIRED_TOKEN",
  "message": "The authorization token provided has expired. Please re-authenticate."
}
```

HTTP Status Code Reference:

HTTP Status	Code String	Description / Cause
200 OK	SUCCESS	Request completed successfully
201 Created	CREATED	New resource created (e.g. Custom Zone)
400 Bad Request	INVALID_PAYLOAD	Malformed JSON body or validation failure
401 Unauthorized	UNAUTHORIZED	Missing or invalid <code>Authorization: Bearer</code> header
403 Forbidden	FORBIDDEN	Caller lacks permissions to access resource
404 Not Found	NOT_FOUND	Requested endpoint or resource ID does not exist
429 Too Many Req	RATE_LIMITED	OTP or API rate limit exceeded
500 Internal Err	SERVER_ERROR	Internal server fault recorded in audit logs

6. Hibir Audit (`audit.hibir.et`) Certification Requirements

To comply with **Hibir Audit (`audit.hibir.et`)** verification guidelines:

1. **No Static Secrets in Code:** Client apps must never send hardcoded API keys. All transactions must be authorized via dynamic Bearer JWT tokens.
 2. **TLS 1.2/1.3 Mandatory Encryption:** All mobile traffic must communicate with `https://api.xdisturb.et:8003`.
 3. **Public Key Hash Pinning:** Pin the server certificate SHA-256 fingerprint in the Dio HTTP client.
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Integration Sign-Off:

- **Lead API Architect:** *Approved*
- **Mobile Integration Lead:** *Approved*
- **Hibir Audit Representative (`audit.hibir.et`):** *Approved*